



What Is **SSH** and Why Did It Replace **Telnet**?



 A BEGINNER-FRIENDLY GUIDE TO
SECURE REMOTE ACCESS



- ✓ ENCRYPTION
- ✓ AUTHENTICATION
- ✓ DATA INTEGRITY

Cisco Router



TOPIC

SSH (Secure Shell)



LEVEL

Beginner



ESTIMATED READING TIME

8 - 10 Minutes



LAST UPDATE

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Lab Included	Yes
Packet Tracer	Yes

Network Fundamentals – Part 1

Introduction

Imagine you're responsible for managing a router located hundreds of kilometers away. One day, a configuration issue occurs, and you need to access the device immediately. Since traveling to the site isn't practical, remote access becomes essential.

For many years, network administrators relied on Telnet to remotely manage network devices. It was simple, widely supported, and easy to configure. However, it had one critical weakness: every piece of information—including usernames and passwords—was transmitted in plain text.

As computer networks expanded and cybersecurity threats became more sophisticated, relying on Telnet was no longer a safe option. The industry needed a secure alternative, and that alternative was Secure Shell (SSH).

In this article, we'll explore what SSH is, how it works, why it replaced Telnet, and why it has become the standard protocol for secure remote administration.

The Problem

Before SSH became the standard for secure remote access, Telnet was one of the most widely used protocols for managing network devices. It allowed administrators to connect to routers, switches, servers, and other devices remotely through a command-line interface (CLI). At the time, this was a significant improvement over having to be physically present at the device.

Despite its simplicity and ease of use, Telnet had a major security flaw. It transmitted all communication—including usernames, passwords, and commands—in plain text. This meant that anyone with access to the network traffic could potentially capture and read sensitive information using a packet analysis tool.

As organizations expanded their networks and remote administration became more common, this weakness became an unacceptable security risk. Protecting administrative access was no longer optional; it became a necessity.

Imagine an administrator logging into a router over Telnet while connected to a public Wi-Fi network. Because Telnet does not encrypt its traffic, an attacker monitoring the same network could potentially capture the administrator's username and password.

The Solution

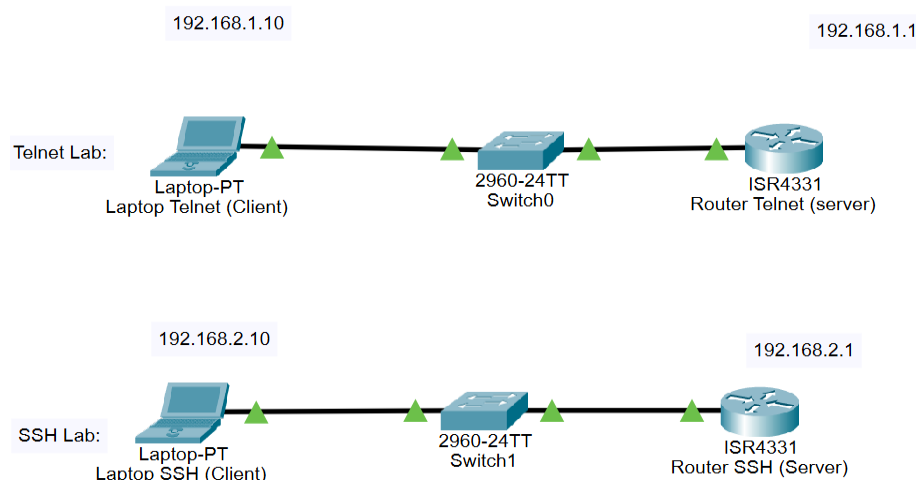
To overcome the security limitations of Telnet, Secure Shell (SSH) was introduced as a secure protocol for remote administration. Unlike Telnet, SSH encrypts all communication between the client and the remote device, protecting sensitive information such as usernames, passwords, and commands from being intercepted.

SSH was designed to provide three essential security features: confidentiality, authentication, and data integrity. These features ensure that remote connections are not only secure but also reliable, making SSH the preferred choice for managing modern network devices and servers.

Today, SSH is widely used by network administrators, system administrators, and cybersecurity professionals to securely access routers, switches, Linux servers, cloud infrastructure, and many other systems.

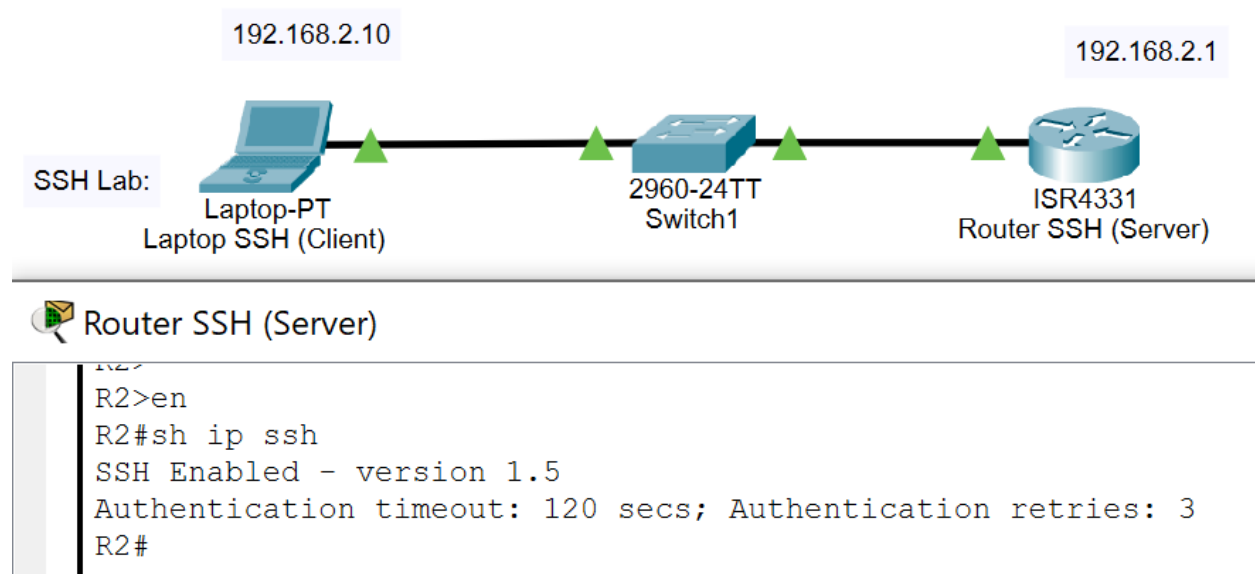
How SSH Works

To better understand how SSH is used in practice, I created a simple Packet Tracer lab comparing Telnet and SSH remote access.



SSH follows a client-server architecture. The client initiates a connection request, while the SSH server listens for incoming connections, typically on TCP port 22.

This lab was implemented in Cisco Packet Tracer, where the simulated IOS reports SSH version 1.5. In production environments, SSH version 2 is recommended due to improved security



When a connection is established, the client first verifies the identity of the server using its host key. This helps prevent users from connecting to an unauthorized or malicious server.

Next, the user is authenticated. This can be done using a password or, more securely, with a public key. Once authentication is successful, SSH creates an encrypted communication channel between the client and the server.

From that point on, every command, response, and file transferred through the SSH session is encrypted, ensuring confidentiality and protecting the communication from unauthorized access.

Advantages of SSH

SSH offers several advantages over traditional remote access protocols such as Telnet. Its primary benefit is encryption, which protects all data exchanged between the client and the server from being read by unauthorized parties.

Another important advantage is strong authentication. SSH supports both password-based authentication and public key authentication, allowing administrators to choose more secure login methods depending on their environment.

SSH also ensures data integrity. This means that information transmitted during a session cannot be modified without being detected, helping to maintain the reliability of communication between devices.

In addition to secure remote administration, SSH supports secure file transfer through protocols such as SCP and SFTP. It also provides features like port forwarding, making it a flexible and powerful tool for system and network administrators.

Real-World Use Cases

Today, SSH is one of the most widely used protocols for secure remote administration. Network administrators use SSH to configure and troubleshoot routers, switches, and firewalls without being physically present at the device.

System administrators rely on SSH to manage Linux servers, perform software updates, monitor system performance, and automate administrative tasks. It is also commonly used to securely access cloud-based virtual machines hosted by providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform.

Because of its security, reliability, and flexibility, SSH has become an essential tool in enterprise networks, data centers, and cloud environments around the world.

Conclusion

Secure remote administration has become a fundamental requirement in modern computer networks. While Telnet played an important role in the early days of networking, its lack of encryption made it unsuitable for today's security requirements.

SSH addressed these limitations by providing encrypted communication, strong authentication, and data integrity. As a result, it has become the industry standard for securely managing network devices, servers, and cloud infrastructure.

Understanding how SSH works is an important step for anyone beginning a journey in networking, system administration, or cybersecurity. Mastering this protocol not only improves technical knowledge but also helps build a strong foundation for more advanced networking concepts.

The complete Packet Tracer lab, configuration files, and supporting resources are available in the accompanying GitHub repository

Learning Note

This article is part of my personal networking learning journey.

My goal is not only to study networking concepts but also to understand them through hands-on practice, document what I learn, and continuously improve my technical skills over time.

As I continue learning, I will publish more articles, build practical labs using Cisco Packet Tracer, and share my progress on GitHub. I hope these resources will also be helpful for others who are beginning their journey in networking and system integration.

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I enjoy building practical labs, writing technical articles, and documenting my learning journey through hands-on projects.

My goal is to continuously improve my technical skills while preparing for a career as a Fachinformatiker für Systemintegration in Germany

